Homework Set 4

Due Date: No Need to Submit

A. TRUE/FALSE

1. TCP/IP makes it possible for a server with IP address X and well-known port Y to simultaneously communicate with multiple remote clients without intermixing response streams.

2. In NFS, a user's modifications to a file must become visible to other users in the system as soon as the write operation completes.

3. A transaction schedule is serializable if and only if the dependency graph (as defined in the lecture) has a cycle.

4. The scheduling $R_1(X), R_2(Y), R_3(Y), W_2(Y), W_1(X), W_3(X), R_2(X), W_2(X)$ for transactions T_1, T_2, T_3 is serializable. (Show the serialization or explain why not.)

B. Short Answers

1. In order for complex data objects to be passed as arguments or results in an RPC over a socket to a possibly remote machine, what needs to be done to the data?

2. The Network File System (NFS) uses a "stateless" protocol between clients and servers. What does this mean? Name one advantage and one disadvantage of stateless filesystem protocols. Can you name one other well-known stateless protocol?

3. For each of the scenarios, state whether you should use UDP or TCP. Briefly explain why.
   1) Streaming live of a lecture
   2) A remote SSH session to test your Project code
   3) A time server broadcasting the correct time to all machines on a LAN

Remote Procedure Calls

For each of the following scenarios, state whether you would use an RPC guaranteeing **at-most-once, at-least-once, or exactly-once semantics** (exactly-once should work in all cases, but if it is overkill you should pick the weakest guarantee that is enough for you application). In some cases multiple answers can be accepted, but you need to make sure your explanation is clear and coherent with your answer, and your (reasonable) assumptions are made explicit – e.g. what application can/must do to avoid exactly once? Explain in no more than 3 sentences.

1) Depositing money into a bank account
2) Uploading your proof of vaccination

3) Making a login attempt online where your account will be frozen after 3 incorrect attempts
4) Cancelling a fraudulent order that's been placed on your credit card
5) Booking a cross-country airplane ticket to go to your internship.

Transaction

Consider the locks for ACID transactions we have discussed in class. We can use read-write lock to implement ACID transaction, and the lock compatibility matrix is as the following:

	ACID-R	ACID-W
ACID-R	√	×
ACID-W	×	×

A recent research paper proposed two extra kinds of transactions in addition to ACID transactions: alkaline transaction and saline transaction1.

- ACID transactions are isolated from all other transactions.
- Alkaline transactions are isolated from other ACID and alkaline transactions.
- Saline transactions expose their intermediate states to all other transactions.
- A saline transaction can include several alkaline transactions.

1 Please fill the Lock Compatibility Matrix below.

	ACID-R	ACID-W	Alka-R	Alka-W	Saline-R	Saline-W
ACID-R	√	×				
ACID-W	×	×				
Alka-R						
Alka-W						
Saline-R						
Saline-W						

2 In addition to being cute[2], what is the benefit of having the two extra types of transactions?

3 To further improve system performance of ACID transactions, Danny proposes that an ACID transaction can release the ACID-R lock immediately after it uses the data. What problem can this proposal cause? Please provide a concrete example.

Distributed File Systems

The TAs have built a distributed file system called Clever File System (CFS). They plan to use CFS to post homework questions, and students can read and update these files with their answers.

[2] [2]English note: 这些名称都是卖萌的叫法，ACID 的意思是酸性，Alkaline 是碱性，saline 是两者结合的，所以叫做盐。

Both clients of TAs and students have a cache, which is empty initially and has a large enough capacity. The cache implements **whole-file caching (like AFS)**, **session semantics**, and **check-on-use**.

Suppose only one TA and one student are accessing a shared directory of CFS (both of them have write access to all files). At first, there is only 1 file named “Q1” in the directory and the file content is: “1+1=?”.

Consider the TA and the student access the files in the following sequences:

Timestamp	TA	Student
1	fd1 = open(“Q1”, "w")	
2	write(fd1, "1*1=?")	
3	fd2 = open(“Q2”, "w")	
4	write(fd2, "2*2=?")	
5	close(fd2)	
6		fd1 = open(“Q1”, "r")
7		read(fd1)
8		close(fd1)
9		fd2 = open(“Q2”, "w")
10	fd3 = open(“Q3”, "w")	
11	write(fd3, "3*3=?")	
12	close(fd3)	
13		write(fd2, "2*2=4")
14	fd2 = open(“Q2”, "rw")	
15	read(fd2)	
16	write(fd2, "2*3=?")	
17	close(fd2)	
18		close(fd2)
19	close(fd1)	
20	fd2 = open(“Q2”, "rw")	
21	read(fd2)	
22	write(fd2, "Hmmm...")	
23	close(fd2)	
24	delete(“Q2”)	

Assume:

- Open operations with the “w” option will create a new file as well as a local cache copy if the file does not exist.
- When the file is opened the offset is 0.
- The client’s cache is large enough that eviction never happens.

- Deleting a file also deletes its local cache copy.

Answer the following questions:

1) Identify which timestamp will cause the client (either TA or student) to fetch file content from the server.

2. Identify which timestamp will cause the client to send file content to the server.

3. What are the read results on timestamp 15 and 21 of TA, and line 7 of student?

4. At the end, what are the file names and contents in TA client cache, student client cache, and the server, respectively?